When Does Filtering Help You See a Break? Latent-State Diagnostics for Structural Change at Calibrated False-Alarm Rates

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Abstract

We ask when filtering a latent state helps detect structural change, using a protocol that calibrates every detector to the same false-alarm rate (5% per 500 observations) on matched null data. The answer is a trichotomy — *no, yes, no* — across three break types. (i) Level shifts in a persistent state: no — a raw-data CUSUM dominates at every SNR, and the innovation CUSUM is provably "fast or never," a boundary condition of persistence ($\mu\infty$ sorts detection, Spearman 0.94). (ii) Observation-noise variance changes: yes, but the advantage is prewhitening, not the state estimate — the observable is exactly ARMA(1,1), so ARIMA and Kalman whitening are provably the same filter. (iii) State-innovation (shock) variance changes — the Great-Moderation/crisis-volatility channel: no, a partial null. Raw matches or beats whitening on the coarse $\times 3$ break (0.72/0.96/0.96 vs 0.26/0.79/1.00), but the subtle $\times 1.5$ break leaves every rung near the false-alarm floor — whitening *fails to recover* it. On real data (industrial production, GDP, Treasury yields, unemployment) every alarm attributes to a second-moment feature; the headline NBER association reaches an uncorrected permutation $p = 0.008$, but this does not survive a family-wise or FDR correction across the full grid of tests the real-data section actually runs (§9), and real-time vintages confirm COVID while downgrading 2008. Two extensions probe the protocol's edges: offline PELT matches raw CUSUM on level breaks but not variance breaks, and a bounded-memory statistic fixes raw CUSUM's blindness to a second level break, though not a second variance break. Read together, the results are deflationary for the latent layer's detection power: a raw or ARIMA-whitened benchmark matches or beats the state-aware detector on every break type studied, and what filtering buys instead is breadth and attribution, not power (§10).

JEL classification: C12, C22, C52. Keywords: structural change; sequential change detection; CUSUM; state-space models; Kalman filter; prewhitening; false-alarm rate; latent-state diagnostics; volatility regimes.

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1. Introduction

Structural change is usually sought in observables. But many economic quantities of interest — trend output, a natural rate, an underlying volatility regime — are latent states observed only with noise, and the natural first step is to filter: estimate the state, then watch the estimate for breaks. Intuition says the filtering should help, by stripping observation noise before the change is sought. This paper asks when that intuition holds, under a protocol that makes the question answerable. Our contributions are as follows.

- Contribution 1 (protocol): a calibrated-FAR parity harness. Every method's threshold comes from the same routine on the same matched-null draws; empirical FAR is re-verified on fresh nulls; causality is enforced structurally (parameters fit on a training prefix only; forward-only filtering; alarm scores are NaN on training data) and tested bit-identically (perturbing future observations leaves all scores at earlier times unchanged, exactly). The protocol extends to the honesty of the research process itself: three pre-registered hypotheses were falsified, and every post-hoc change is logged (CHANGELOG) and the failures reported as findings rather than suppressed.
- Contribution 2 (negative + positive results): the intuition is wrong for first moments at high persistence — and we can prove why (fast-or-never theorem), then bound the claim by sweeping φ ($\mu\infty$ sorts detection, and the innovation CUSUM escapes the regime at low φ). For second moments the answer is a two-rung, two-channel decomposition. The ladder collapses to *two* rungs because the ARIMA and Kalman rungs are the same filter — the observable is exactly ARMA(1,1), an equivalence we state as theory and confirm to machine precision — so "prewhitening" and "state estimation" are not separable here; the state layer's channel-specific advantage on this DGP reduces to prewhitening, which ARIMA residuals also supply. And the prewhitening advantage is *channel-specific*: it holds for observation-noise breaks (decisive at high SNR, where the latent signal masks the noise change and raw sits at chance) but *inverts* for state-innovation breaks, where the raw variance CUSUM matches or beats the whitened rungs — the channel that matters for the volatility-regime events of the application.
- Contribution 3 (method): a tail-robust exceedance-indicator CUSUM that preserves the second-moment advantage under heavy-tailed noise, reached by rejecting two documented alternatives — a falsified clipped variant and an in-composite variant diluted below its standalone form (§8.3).
- Contribution 4 (application discipline): attribution, permutation tests, sensitivity, pinned data snapshots, and real-time vintages for the real-data claims — including a self-correction on the headline GFC timing.

Related work. The paper sits at the intersection of several literatures.

Quickest detection and SPC. Sequential change detection descends from Page's (1954) CUSUM and the quickest-detection tradition (Lorden 1971; Moustakides 1986); we use the CUSUM as the common statistic across information sets rather than proposing a new stopping rule. That literature also supplies the evaluation currency we adopt: the average run length (ARL), with methods compared at a matched in-control ARL_0 — the statistical-process-control convention (Page 1954; Montgomery 2013). The broader statistical-surveillance research program this evaluation convention sits inside is surveyed in Frisén (2003), a complement to the Basseville & Nikiforov (1993) treatment cited below for the innovation-CUSUM axis specifically.

Innovation-based state-space monitoring. The idea of monitoring a *state-space* model through its innovations is not ours: it is the core of the innovation-based change-detection literature — the generalized likelihood ratio test on Kalman innovations (Willsky & Jones 1976) and the systematic treatment of CUSUM/GLR schemes on innovation sequences in Basseville & Nikiforov (1993). Our contribution on this axis is *not* the innovation CUSUM or ARL-matching per se, both of which are standard in SPC and quickest detection; it is that this matched-error protocol is essentially *absent* from the applied latent-state econometrics literature, and that transplanting it there — calibrating every information set to a common false-alarm rate — is precisely what produces the negative results (raw data wins on levels; the state estimate adds nothing over ARIMA whitening on variances).

Econometric CUSUM-of-residuals. In econometrics, sequential monitoring of structural change is the CUSUM-of-recursive-residuals line and its modern moving-window ("MOSUM") monitoring form (Brown, Durbin & Evans 1975; Chu, Stinchcombe & White 1996), which watch *observable* regression residuals; our object is a *latent* state, and the residuals we monitor are a filter's innovations. The bounded-memory fix we give the multi-break re-arm failure in §7 is in this same MOSUM family: a moving two-window mean-shift statistic, rather than a CUSUM accumulated against a fixed training-prefix baseline.

Regime-switching. Regime-switching models (Hamilton 1989; Kim & Nelson 1999) offer a state-aware alternative whose regime probabilities we include as a benchmark and find saturate under calibration on nonstationary data.

Great Moderation and volatility empirics. The empirical target of the second-moment results is the Great Moderation volatility decline (McConnell & Pérez-Quirós 2000; Stock & Watson 2002); as we show, that and the crisis-volatility events are *state-innovation* (shock-variance) breaks, the channel on which raw and prewhitened detectors are interchangeable.

Parametric volatility models. A separate literature models conditional variance parametrically — GARCH (Bollerslev 1986) and stochastic-volatility state-space models (Kim, Shephard & Chib 1998); its object is *estimation* of a volatility process, not distribution-free monitoring at a calibrated false-alarm rate, so a full stochastic-volatility comparison remains future work. A GARCH(1,1) benchmark is reportable now, however: fit on the training prefix only (via the `arch` package), causally forward-filtered for conditional variance over the full series with the fixed fitted parameters, and run through the same three-arm max-CUSUM used for the raw and ARIMA rungs on its standardized residuals (`experiments/garch_detector.py`, `experiments/exp15_garch_benchmark.py`). Calibrated at n_reps = 500 matching the published grid (empirical FAR 0.050 in every cell checked), GARCH sits at the false-alarm floor on both variance channels at every SNR tested — r-channel ×1.5: 0.098 (SNR 0.5), 0.096 (SNR 2.0); q-channel ×1.5: 0.066 (SNR 0.5), 0.098 (SNR 2.0) — against raw's 0.56/0.10/0.21/0.23 and ARIMA's 0.94/0.87/0.10/0.16 on the same four cells (§5, Table 3). This is not merely "GARCH is dominated": on this DGP it contributes nothing over chance. The likely mechanism is a generative mismatch — GARCH(1,1) is built for conditional heteroskedasticity (volatility clustering driven by squared past shocks), a different assumption than this paper's DGP (a permanent step change in noise variance layered on a highly persistent $\varphi = 0.95$ latent state) — but we have not isolated the mechanism beyond this scope note, and do not rule out an implementation-specific cause. A break-aware GARCH variant (allowing its own parameters to shift) and the full stochastic-volatility comparison remain open.

Offline changepoint detection. Offline changepoint methods (PELT, Killick et al. 2012) solve a retrospective segmentation problem; our monitoring is strictly causal and calibrated to a false-alarm rate, so the two are not comparable on delay (§8.5). We nonetheless calibrate PELT to the same false-alarm rate on null paths and ask whether it *localizes* a break at all, given the full sample: it is competitive with the raw-Y CUSUM on an obvious 3σ level shift (localization rate 0.83–0.92 across SNRs vs. the causal detector's 0.97–0.99) but far weaker on pure variance breaks (0.00–0.02 vs. the dedicated raw variance-CUSUM's 0.10–1.00 across the same grid), because PELT's default cost model is a mean-shift detector — a concrete reason a dedicated variance statistic, not an off-the-shelf offline method, is the right tool for the channel this paper is about. The canonical retrospective alternative for multiple mean breaks specifically, rather than PELT's general segmentation cost, is the dynamic-programming estimator of Bai & Perron (2003); we do not benchmark against it directly since our question is causal monitoring at a calibrated false-alarm rate, not retrospective break-date estimation, but it is the natural reference point for §7's multi-break discussion.

What is new here is not any single detector but the calibrated-parity harness that makes latent-state and raw-data detectors directly comparable, and a reduced-form result (the exact ARMA(1,1) equivalence of the ARIMA and Kalman rungs) that dissolves the "does filtering help?" question for second moments into a prewhitening question with a known answer.

2. Framework and evaluation protocol

Model layer. S_t = φS_{t−1} + w_t (var q), Y_t = S_t + v_t (var r); the estimator sees only Y and fits (φ, q, r) by maximum likelihood on a training prefix of 125 observations (25% of the T = 500 baseline sample; the same 25% fraction at other T), then runs a forward-only filter with frozen parameters. Standardized one-step innovations e_t and the filtered state are the raw material for diagnostics.

Diagnostics layer. Eleven features of the filtered path — level change, slope, acceleration, instability, rolling persistence, a Page CUSUM of innovations (allowance k = 0.5), variance CUSUMs of e^2-1 (k = 0.25 and 0.05), a quietness CUSUM of $1-e^2$ (k = 0.05), rolling innovation autocorrelation, and a CUSUM of the filtered state against its training baseline. Each feature is standardized at every time point by the median and IQR of its null distribution at that same time point (pooled-over-time standardization is a design flaw that blunted the composite; §8.4). The composite score is the max of standardized features; the composite is itself calibrated on nulls, which prices in the multiple-feature testing automatically.

Calibration parity. Threshold = (1−FAR) quantile of the per-replication maximum score over matched-null draws; identical routine, budgets, and seed layout for LSC detectors and benchmarks (raw-Y CUSUM, ARIMA+CUSUM, plain-HMM regime flips). Calibration, evaluation, FAR-check, and feature-scale seeds are disjoint by construction.

Metrics. Detection rate, pre-break false alarms, censored delay; for multi-break: event-level precision/recall/F1 with one-to-one greedy matching in a 100-observation window. To connect the protocol to the statistical-process-control and quickest-detection literatures we also report the two average-run-length quantities those fields are built around (`paper_assets/arl_table.csv`). The in-control ARL_0 is the mean observations between false alarms implied by the calibrated per-observation false-alarm hazard: with monitored window length L = 375 (T = 500, 25% training) and empirical window-FAR α, the hazard is p = 1 − (1−α)^{1/L} and $ARL_0 = 1/p \approx L/\alpha$; our 5% window target corresponds to $ARL_0 \approx 7300$ observations, and the empirical detectors sit at 5900–7600 (raw CUSUM calibrates hot, increasingly so at higher SNR: 4.0% → 6.2% → 8.2% vs. the 5% target). The out-of-control ARL_1 is the post-break detection delay (mean delay conditional on detection, reported beside the detection rate since misses are censored): e.g. at 3σ level, SNR 0.5, $ARL_1 \approx 77$–86 observations. Calibrating a common ARL_0 is exactly the ARL-matching convention of SPC (Basseville & Nikiforov 1993); we express it as a window-FAR because our horizon is finite.

Arena	Method	Empirical FAR	ARL_0 (obs)
SNR 0.1	lsc_composite	5.2%	7023
SNR 0.1	lsc_kalman_cusum	3.4%	10841
SNR 0.1	lsc_state_cusum	5.0%	7311
SNR 0.1	raw_cusum	4.0%	9187
SNR 0.5	lsc_composite	4.8%	7624
SNR 0.5	lsc_kalman_cusum	4.8%	7624
SNR 0.5	lsc_state_cusum	5.4%	6756
SNR 0.5	raw_cusum	6.2%	5859
SNR 2.0	lsc_composite	6.2%	5859
SNR 2.0	lsc_kalman_cusum	4.8%	7624
SNR 2.0	lsc_state_cusum	7.8%	4618
SNR 2.0	raw_cusum	8.2%	4383

Table 1. ARL_0 by arena and method (grid_v1 core, T = 500; target $ARL_0 \approx 7300$ at the 5% window-FAR).

MC SEs on empirical FAR $\leq$ 0.013 (n_reps = 500, $\sqrt{(p(1-p)/n\_reps)}$).

Arena	Scenario	Method	Detect rate	ARL_1 (mean delay)
SNR 0.1	level 3σ	lsc_composite	0.766	112.6
SNR 0.1	level 3σ	lsc_kalman_cusum	0.654	63.9
SNR 0.1	level 3σ	raw_cusum	0.966	72.3
SNR 0.1	variance ×3	lsc_composite	0.990	29.3
SNR 0.1	variance ×3	lsc_kalman_cusum	0.870	80.4
SNR 0.1	variance ×3	raw_cusum	0.728	104.0
SNR 0.5	level 3σ	lsc_composite	0.530	86.4
SNR 0.5	level 3σ	lsc_kalman_cusum	0.554	77.2
SNR 0.5	level 3σ	raw_cusum	0.990	81.8
SNR 0.5	variance ×3	lsc_composite	0.992	25.2
SNR 0.5	variance ×3	lsc_kalman_cusum	0.224	107.2
SNR 0.5	variance ×3	raw_cusum	0.076	137.6
SNR 2.0	level 3σ	lsc_composite	0.670	63.2
SNR 2.0	level 3σ	lsc_kalman_cusum	0.674	49.0
SNR 2.0	level 3σ	raw_cusum	0.988	96.6
SNR 2.0	variance ×3	lsc_composite	0.976	17.2
SNR 2.0	variance ×3	lsc_kalman_cusum	0.268	114.6
SNR 2.0	variance ×3	raw_cusum	0.058	172.9

Table 2. ARL_1 (detection rate and mean delay conditional on detection) at the canonical level-3σ and variance-×3 breaks, T = 500. MC SEs on detect rate $\leq$ 0.023 (n_reps = 500).

3. Simulation design

Arenas: AR(1) latent state (φ = 0.95) at spec-SNR (stationary state variance / observation variance) ∈ {0.1, 0.5, 2.0}, with a persistence sweep φ ∈ {0.5, 0.8, 0.95, 0.99} (SNR held fixed by $q = SNR\cdot(1-\varphi^2)$) used to map the fast-or-never boundary (§4); and a local-level (random-walk state) arena, analyzed in §8.4, where — as we now demonstrate rather than assert — level detection is degenerate for every method and variance detection becomes possible only after whitening. Breaks at mid-sample come in two variance channels whose distinction is central to §5. An *observation-noise* (r) break scales the white-noise component's standard deviation (×1.5, ×3, ×⅔ quieting); a *state-innovation* (q) break scales the state shock's standard deviation by the same factors — the identical SD convention, so "×1.5" means SD×1.5 in both channels. The two are structurally different: an r-break changes only the marginal variance of Y, whereas a q-break changes both the marginal variance and the autocorrelation structure (it shifts the reduced-form ARMA(1,1) MA parameter). Also: level shifts (0.5, 1, 3 σ_ref, where σ_ref = $\sqrt{(q/(1-\varphi^2))}$ is the latent state's own stationary standard deviation), logistic ramps, and pure persistence changes (φ → 0.995 or 0.80 with the stationary variance held fixed). T = 500 baseline with a T ∈ {200, 2000} sweep; misspecification arenas with t_5 observation noise and a nonlinear tanh state drift.

4. First moments: raw data wins, and we can say exactly why

Empirics. At matched FAR, raw-Y CUSUM has the best level-shift detection rate at every SNR (0.97–0.99 at 3σ) and every T; the latent state CUSUM approaches but never overtakes it (0.19 → 0.94 as SNR rises). Two pre-registered "latent advantage" hypotheses were falsified. The latent-innovation CUSUM is the speed champion conditional on firing: median delay 24–53 observations at 3σ versus raw's 58–91, at the cost of detection rate (0.55–0.67) — "fast or never."

Theory. The empirics above are not an artifact of tuning; they follow from two facts about the detectors' post-break drift. Statements and proofs are in Appendix B (long-form companion: `experiments/THEORY.md`).

Proposition 1 (fast-or-never). After a state level shift δ, the standardized innovation mean decays geometrically at rate $\rho = \varphi(1-K)$ to $\mu\infty = \delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F})$, where K is the steady-state Kalman gain and F the innovation variance. If $\mu\infty < k$, the post-transient innovation CUSUM has negative drift, and the probability of an alarm in the next L observations is bounded by $(L+1)\cdot\exp(-2(k-\mu\infty)h)$ for threshold h.

The detector therefore fires during the adaptation transient or, with exponentially small probability, never. With $\varphi = 0.95$ and $k = 0.5$, $\mu\infty > k$ would require shifts of order 10σ: the innovation CUSUM is *structurally* in the fast-or-never regime, which is why it is the speed champion but never the detection champion.

Fast-or-never is a property of the whitening filter, not of Kalman. By the ARMA(1,1) equivalence (§5, Appendix B) the innovations here are the innovations of the reduced-form ARMA(1,1) of Y, so Proposition 1 can be stated with no state space at all: after the shift, the standardized ARMA(1,1) innovation mean decays geometrically *at rate equal to the MA parameter* $\theta = \varphi(1 - K)$ to the same $\mu\infty = \delta(1-\varphi)/((1-\theta)\sqrt{\sigma\_\varepsilon^2})$, and the bound applies verbatim. Two consequences: an ARIMA(1,0,1) residual CUSUM inherits the *same* fast-or-never behavior as the Kalman-innovation CUSUM (they whiten to the same series — nothing is Kalman-specific), and the decay rate θ is *observable* — an analyst reads it off a fitted ARMA(1,1) without positing a latent state. Since $\theta \to 1$ as $\varphi \to 1$, the trap is a persistence phenomenon, which is what the φ sweep tests next.

Proposition 2 (raw-CUSUM delay). The raw-Y CUSUM sees the full shift as a sustained standardized drift $\Delta = \delta/\sigma\_Y$, giving a Wald first-passage mean delay of approximately $h/(\Delta-k)$ once $\Delta > k$ (Wald 1947; Siegmund 1985), and negligible power when $\Delta \leq k$.

Verification (`experiments/exp06_theory_check.py`, 1000 reps): the Monte Carlo innovation path matches μ_t within MC error; the Proposition 1 bound is never violated; at the actual calibrated thresholds, $\mu\infty$ sorts every observed detection rate — $\delta \leq 1\sigma$ gives bound $\leq 0.7\%$ (observed $\approx$ FAR); $\delta = 3\sigma$ is a knife-edge ($\mu\infty = 0.43$–0.48 vs $k = 0.5$) matching the observed 0.55–0.67. Proposition 2's Wald delays 68/84/110 across SNRs bracket the observed raw medians 58/75/91 ($\approx$ 15–20% conservative) and explain the partial 1σ detection (0.30) without any fitting: $\Delta = 0.577$ barely exceeds k, so the Wald delay (1334) dwarfs the 250-observation horizon.

The persistence boundary (φ sweep). Proposition 1's regime is not unconditional — $\mu\infty = \delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F})$ is *increasing in* $(1-\varphi)$, so the fast-or-never trap should loosen as the state becomes less persistent. Sweeping $\varphi \in \{0.5, 0.8, 0.95, 0.99\}$ at fixed SNR (Figure 1) confirms it and turns the theory into a falsifiable ordering: $\mu\infty$ sorts the innovation-CUSUM detection rate across all cells (Spearman 0.94, n=24: 4 φ × 3 SNR × 2 shifts — a stratified permutation null that shuffles φ-pairing within each SNR×shift stratum, so only φ/μ∞ specifically, not SNR or shift size, can drive the ordering, rejects at $p<0.00005$ over 20,000 draws; `experiments/exp12_spearman_null_test.py`), fast-regime cells ($\mu\infty \geq k = 0.5$) detect 0.83–1.00 while never-regime cells ($\mu\infty < 0.5$) detect 0.07–0.67, and at 3σ the detector escapes the trap at low persistence (0.98/0.97 at $\varphi = 0.5/0.8$) but is caught at high persistence (0.65/0.30 at $\varphi = 0.95/0.99$) — while the raw CUSUM detects at 0.96–1.00 for *every* φ, so the φ-dependence is specific to the filtered detector. The negative first-moment result is therefore a boundary condition of *persistent* latent states — which is the empirically relevant case (trend output, a natural rate, a volatility level are near-unit-root), and the case the rest of the paper studies at $\varphi = 0.95$. The one off-trend cell — $\varphi = 0.99$, low SNR, where detection (0.63) exceeds what $\mu\infty$ (0.21) predicts — is honest evidence of the transient's own contribution: at near-unit-root φ the adaptation transient is so long that its accumulated mass fires the CUSUM even when the asymptotic drift is negligible (the

”fast” branch, strengthened by a long transient). $\mu\infty$ governs the post-transient tail; total detection adds the transient mass.

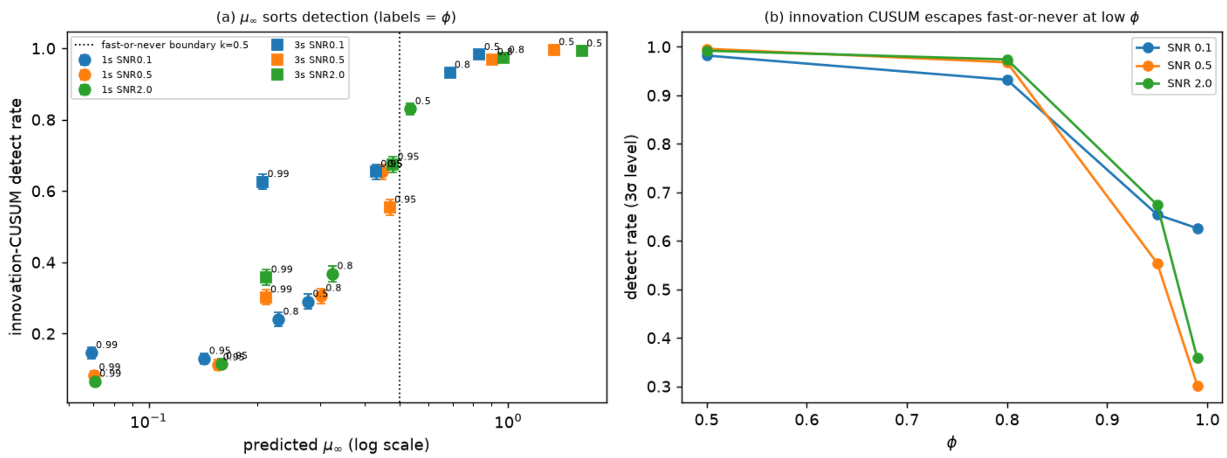

Figure 1. The φ sweep (`grid_v6_phisweep`). Left: predicted asymptotic innovation drift $\mu\infty$ against the observed innovation-CUSUM detection rate for every $\varphi \times$ SNR $\times$ shift cell (point labels = φ); $\mu\infty$ sorts detection (Spearman 0.94), and the dotted line marks the fast-or-never boundary k = 0.5. Right: at 3σ the detector escapes the trap at low persistence (0.97–1.00 at $\varphi = 0.5$–0.8) and is caught at high persistence (0.30–0.67 at $\varphi = 0.95$–0.99).

Assumptions and estimation error. Propositions 1–2 assume the steady-state filter with *known* parameters. Two facts keep this from being a limitation in practice. First, the filter reaches its steady state within the 125-observation training prefix of §2 (25% of T = 500), well before monitoring begins (the innovation autocorrelation is flat by then). Second, the error from estimating (φ, q, r) rather than knowing them is second-order: in the M1 equivalence check (§5, `experiments/exp07`) the ARIMA and Kalman standardized-innovation series computed with *estimated* parameters correlate at $\bar{\rho} = 0.99$ with each other and, computed with *true* parameters, at $\rho = 1.000$ to a max discrepancy of $\approx 10^{-9}$ — so the entire estimated-vs-true gap is the small residual that leaves the median correlation at 0.99. The theory describes the estimated filter to within that gap. The scope of this claim should be stated precisely: $\bar{\rho} = 0.99$ establishes agreement of the innovation *series* under estimation; the detection-rate consequences of estimation error are shown empirically — every grid in the paper is run with estimated parameters — not proved analytically. On at least one margin this gap is not second-order: for the Table 2 flagship cell (SNR 0.5, φ = 0.95, level 3σ), holding sidedness and calibration fixed and replacing estimated (φ, q, r) with the arena's true values raises the two-sided innovation-CUSUM detection rate from 0.554 to 0.970, and pairing known parameters with a one-sided CUSUM (a common alternative convention, not what Table 2 reports) reaches 0.990 — so a reader reimplementing this cell under either convention should expect a substantially higher detection rate than 0.554, which reflects the estimated-parameter, two-sided construction actually used, not an error (`experiments/exp10_cusum_ablation.py`).

A dense magnitude continuum, not just the reported points. The level-shift results above are demonstrated at 0.5/1/3 σ_ref; a referee-requested check reruns the actual raw and innovation CUSUM detectors (`make_raw_cusum_detector`, `make_innovation_cusum_detector`), with estimated (training-prefix MLE) parameters as in Table 2, across a dense 21-point magnitude grid (0.0–4.0 σ_ref in steps of 0.2) at the three benchmark SNRs, φ = 0.95, T = 500 (`experiments/exp11_break_magnitude_sweep.py`, `paper_assets/exp11_level_sweep.csv`). Raw's dominance holds at every one of the 63 grid points (0 violations where raw trails innovation by more than 3pp) — not just the two or three magnitudes reported in Table 2. The knife-edge framing needs one qualification the discrete points did not surface: the *empirical* ratio at which the innovation CUSUM's detection rate crosses 50% (ratio $\approx$ 2.0–2.4 across the three SNRs) sits well below the ratio at which the *theoretical* $\mu\infty$ crosses $k = 0.5$ (ratio $\approx$ 3.1–3.5) — consistent with the transient-mass effect already noted for the $\varphi = 0.99$ low-SNR cell above, now shown to be a general, continuous phenomenon rather than a single favorable cell: $\mu\infty$ still sorts detection (the φ-sweep Spearman result), but is not itself the empirical half-detection point.

5. Second moments: a whitening ladder

Is the latent layer's second-moment advantage about the *state estimate*, or merely about *prewhitening* the autocorrelated observations before applying a variance statistic? We probe this with a three-rung ladder, calibrated by the same routine on the same matched-null seed blocks:

- raw — a variance CUSUM (up-arm Page CUSUMs of z^2-1 at allowances $k = 0.25$ and 0.05, a down-arm CUSUM of $1-z^2$, max over arms, no per-time standardization) on z from the raw observations, standardized by frozen training-prefix moments;
- ARIMA — the identical statistic on the standardized one-step residuals of an AIC-selected, training-prefix-frozen ARIMA model (whitened, but not state-aware);
- latent — not the same statistic re-run on Kalman innovations, but the paper's eleven-feature diagnostic composite (§2) evaluated on the Kalman-filtered path, whose score is a max over features each standardized per time point against Monte Carlo null replications.

The raw and ARIMA rungs isolate one axis — prewhitening — holding the detection statistic fixed. The latent rung changes two things at once, the information set (Kalman-filtered, state-aware) and the statistic itself (composite, per-time-standardized, versus the lower rungs' unstandardized 3-arm CUSUM), so its comparison to raw/ARIMA below should be read as *state-aware composite* vs. *whitened single statistic*, not as a third setting of one controlled instrument.

The ladder has three rungs but only two are distinct. For the AR(1)

- noise DGP the observable Y has an *exact* ARMA(1,1) reduced form: differencing by $(1 - \varphi L)$ leaves an MA(1), whose invertible root gives an MA parameter θ and innovation variance σ_ε^2 satisfying two identities we verify to machine precision (Appendix B; `lsc.theory .arma11_representation`) — $\sigma\_\varepsilon^2 = F$, the Kalman innovation variance, and $\theta = \rho = \varphi(1 - K)$, the Proposition-1 decay rate. So the steady-state Kalman innovations *are* the ARMA(1,1) innovations: the ARIMA and latent rungs are the same filter, not two competitors. We confirm this numerically (`experiments/exp07_arma_equivalence.py`, ≥ 200 null paths): with true parameters the two standardized-innovation series agree to a median correlation of 1.000 (max discrepancy $\approx 10^{-9}$), and with each rung fit on the training prefix — the actual operating condition — they still correlate at $\bar{\rho} = 0.99$, the small wedge being pure estimation error (forcing the ARIMA order to the true (1,0,1) tightens it to 0.9995). (AIC, incidentally, rarely selects (1,0,1): near the unit root at $\varphi = 0.95$ it prefers (1,0,0) or a differencing (0,1,1); this is a benign near-unit-root artifact — those orders approximate the ARMA(1,1) closely enough to preserve $\bar{\rho} \geq 0.95$ — reported in full in Appendix B.) The practical consequence: the ladder is really raw vs. whitened, and "does the *state estimate* help beyond ARIMA whitening?" has the answer *no, by construction*. What remains is the genuinely empirical question — when does whitening help at all? — which turns out to depend on *which variance channel* breaks.

Table 3. The ladder, both channels (detection rate at T = 500, 5% calibrated FAR; MC SEs ≤ 0.02 in `paper_assets/ladder_table.csv`, `break_channel` column).

channel	break	rung	SNR 0.1	SNR 0.5	SNR 2.0
r (obs-noise)	×1.5	raw	1.00	0.56	0.10
		ARIMA	0.90	0.94	0.87
		latent	0.82	0.87	0.91
	×3	raw	1.00	1.00	0.85
		ARIMA	0.98	1.00	1.00
		latent	0.99	0.99	0.98
q (state-innov)	×1.5	raw	0.09	0.21	0.23
		ARIMA	0.03	0.10	0.16
		latent	0.06	0.11	0.23
	×3	raw	0.72	0.96	0.96
		ARIMA	0.26	0.79	1.00
		latent	0.44	0.76	0.98

Reading the ladder: the ordering inverts across channels. The subtle ×1.5 break is the discriminating case, and it tells opposite stories in the two channels.

Observation-noise (r) breaks — prewhitening wins. The *raw* rung is strongly SNR-dependent, falling monotonically from 0.996 (SNR 0.1) through 0.560 to 0.102 (SNR 2.0, within 5 pp of the 6.0% empirical FAR, i.e. chance). The mechanism is transparent: as SNR rises the latent state's variance dominates the marginal variance of Y, so a ×1.5 change in the (now-small) observation-noise component is a shrinking fraction of total variance, masked by state-driven autocorrelation. Prewhitening removes exactly that autocorrelation: the *ARIMA* rung is flat across SNR (0.90 / 0.94 / 0.87) and the *latent* rung tracks it closely (0.82 / 0.87 / 0.91) — an empirical proximity, not a consequence of the equivalence above, which concerns the shared innovation series rather than the composite statistic built on top of it. On this channel the advantage over raw is *prewhitening under autocorrelation*, decisive where the latent signal masks the noise change.

State-innovation (q) breaks — the ordering inverts, read separately by break size. The *coarse* ×3 break carries the inversion claim cleanly: every rung detects above chance, and the raw rung matches or beats the whitened rungs at every SNR (raw 0.72 / 0.96 / 0.96 vs ARIMA 0.26 / 0.79 / 1.00, the whitened rung catching up only at the high-SNR ceiling). The *subtle* ×1.5 break must be read against the same $\leq$ 5-pp-from-FAR "chance" standard applied to the r channel above (empirical FARs on this grid are 4.2–6.6%): raw at SNR 0.1 (0.09) and ARIMA at SNR 0.1 and 0.5 (0.03, 0.10) are within 5 pp of their empirical FARs — chance — and only the higher-SNR cells clear the standard (raw 0.21 / 0.23 at SNR 0.5 / 2.0; ARIMA 0.16 at SNR 2.0). The honest subtle-break statement is therefore not "raw detects it" but "whitening fails to recover it": no rung detects a ×1.5 shock-variance break well, and whitening only loses ground. The raw rung's SNR-dependence still *reverses sign* relative to the r channel — it rises with SNR ($0.09 \rightarrow 0.21 \rightarrow 0.23$) — and the mechanism is the mirror image: a q-break inflates the *state's own* variance, which dominates the marginal variance of Y at high SNR, so a raw z^2 statistic sees it directly — whereas prewhitening *strips out* the state-carried signal along with the autocorrelation. Prewhitening reveals a break in the white component and removes a break in the state. Quieting (×⅔, i.e. reduced q) is undetectable by every rung (≤ 0.07, at FAR): a low-q state contributes too little to Y to register its own reduction.

Where does raw's q-break advantage come from? A φ sweep. Holding the shock variance q and observation variance r fixed and sweeping the persistence φ makes the state's stationary variance $q/(1-\varphi^2)$ — and hence the induced SNR — rise as the $1/(1-\varphi^2)$ amplification factor (Figure 2; `configs/grid_v8_phiqbreak.yaml`). The result is instructive and only partly what one might guess. On the *subtle* ×1.5 break, raw's advantage Δ = detect(raw) − detect(ARIMA) is driven by exactly this amplification: it vanishes when the state is white (Δ = 0.00 at φ = 0.1, where whitening is a no-op), rises with the amplification (Spearman 0.83), and — the clean check — equals the SNR-sweep value from the fixed-φ grid at the *same induced SNR* (Δ = 0.11 vs 0.11 at SNR 0.5; 0.07 vs 0.07 at SNR ≈ 2), so the φ-sweep and the SNR-sweep are one experiment: raw's edge on a subtle shock-variance break is the state's variance share, nothing more. (These Δ values are differences between near-floor detection rates — e.g. 0.21 vs 0.10 at SNR 0.5 — so they measure the visibility of a marginal advantage, not a large power gap.) But the law is not monotone to the unit root — Δ peaks at φ = 0.95 and recedes at φ = 0.99, because there the raw detector's own baseline degrades (its calibrated threshold jumps from 275 to 1829, the nonstationarity penalty of §8.4). And on the *coarse* ×3 break the amplification story does not hold at all: raw beats ARIMA at every φ, including φ = 0.1 (Δ = 0.34), with Spearman(amp, Δ) negative — a gross shock-variance break is visible to raw z^2 regardless of amplification, while whitening removes it. (This is the whitening itself, not order mis-selection: at low φ the ARIMA rung selects a stationary AR(1), not a differencing model.) The honest reading: prewhitening strips the state-carried variance signal for q-breaks of any size — that is B2 — and the $1/(1-\varphi^2)$ amplification governs only *how visible* the residual raw advantage is on the marginal, subtle break.

φ	Amplification $1/(1-\varphi^2)$	Δ, subtle ×1.5 (raw − ARIMA)	Δ, coarse ×3 (raw − ARIMA)
0.10	1.01	0.000	0.344
0.50	1.33	0.016	0.526
0.70	1.96	0.038	0.528
0.85	3.60	0.104	0.204
0.95	10.26	0.112	0.168
0.99	50.25	0.074	0.302

Table 4. Raw's detection-rate advantage over the ARIMA rung (Δ), swept over φ at fixed q, r (`grid_v8_phiqbreak`). On the subtle break Δ tracks the amplification factor and peaks at φ = 0.95 before receding at the unit-root edge; on the coarse break Δ stays large at every φ, including φ = 0.1 where amplification is negligible.

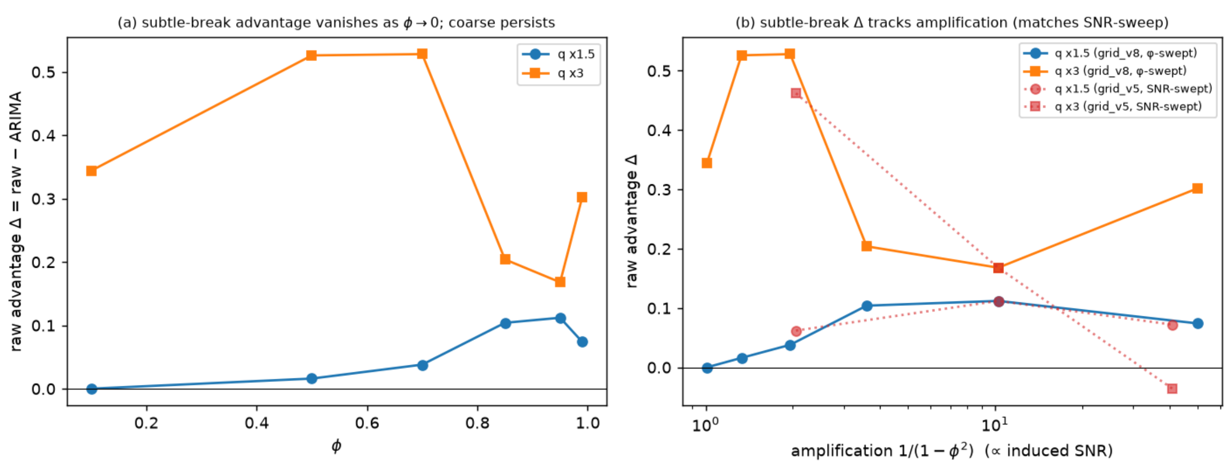

Figure 2. The φ × q cross-grid (`grid_v8_phiqbreak`). Left: raw's advantage Δ = detect(raw) − detect(ARIMA) against φ; the subtle ×1.5 advantage vanishes at φ = 0.1 and peaks at φ = 0.95, while the coarse ×3 advantage persists at every φ. Right: the same Δ against the $1/(1-\varphi^2)$ amplification factor, with the fixed-φ SNR-sweep values (grid_v5) overlaid — on the subtle break the φ-swept and SNR-swept experiments coincide at matched induced SNR; note the Δ values are differences between near-floor detection rates.

This resolves the pre-registered decision rule (`experiments/CHANGELOG.md`) as Outcome B2: the "prewhitening beats raw" result is *specific to the observation-noise channel*. It matters because the events the application targets — the Great Moderation, crisis volatility — are state-innovation (shock-variance) breaks, not observation-noise breaks; on that channel a raw variance CUSUM is at least as good as any amount of whitening — decisively on the coarse break, while on the subtle break every rung is near the floor and whitening only loses ground. This is not a weakness to hide but the explanation for a real-data fact (§9): the raw variance CUSUM's crisis timing is *indistinguishable* from the state-aware composite's, exactly as the q- channel predicts. (The earlier r-channel-only framing recorded a provisional "Outcome C"; the q channel is what disambiguates it.) Attribution: the z^2/e^2 variance-pressure arms drive the alarms at every rung.

Runway (T sweep, ×1.5, SNR 0.5). The subtle-break detection scales with sample length for every rung: at T = 200 all three are weak (raw 0.26, ARIMA 0.10, latent composite 0.11) — the whitened rungs need runway to accumulate the CUSUM — and by T = 2000 all reach ceiling (raw 0.98, ARIMA ≈ 1.00, latent 0.99). The T = 200 calibration is *not* hot for these variance detectors (empirical FAR 5.0–5.8% vs the 5% target), unlike the level CUSUMs (§8.1). At the coarse ×3 break the composite is at ceiling even at T = 200 (1.00, median delay 26 of the ~100 post-break observations, best level-oriented benchmark 0.17).

Scale *quieting* (×⅔) inverts the up-break pattern and is treated in §6 and §8.3: there the raw rung catches the low-SNR case (0.32 at SNR 0.1, where noise dominates) while the whitened rungs and composite sit at chance, and only the standalone exceedance detector recovers it at moderate SNR.

6. Dynamics: near the information floor

Pure persistence changes (marginals preserved) are close to undetectable: at SNR 0.5, no method exceeds FAR (a pre-registered hypothesis wrong in both directions — raw CUSUM even reached 0.16 on persistence-up via a conditional "level-freeze" artifact we dissect). Quieting changes (φ down) actively *suppress* every excursion statistic below its null level. Purpose-built quietness features (CUSUM of $1-e^2$, rolling innovation autocorrelation) rescue detection only where the information exists: 0.33 at SNR 2.0 (the only above-FAR persistence detection anywhere in the grid) and 0.17 at T = 2000. That 0.33 is a single favorable cell, not a broad capability: its value is that it maps where the information floor sits, showing detection is possible exactly where the state is most visible and nowhere else. Scale-quieting (×⅔) is detectable — but, outside the noise-dominated regime, only by the exceedance detector (§8.3): at SNR 0.5 it reaches 0.41/0.33 (Gaussian/t_5) while every other method sits at chance. The one exception is the same low-SNR window that lifts the raw rung on the up-breaks (§5): at SNR 0.1 the raw variance CUSUM catches the quieting too (0.32), because when observation noise dominates a scale change is directly visible in raw z^2 regardless of its sign.

7. Multiple breaks: everyone is one-shot

Under a re-arm protocol applied identically to all methods (re-arm when the score drains below half threshold plus a 20-obs refractory), raw CUSUM never detects a second event (recall 0.00 — its fixed-baseline statistic saturates and cannot drain), but the LSC CUSUMs rarely re-arm in time either at 150-observation spacing (level→level second-event recall ≤ 0.05): first-alarm delay plus drain time exceeds the gap. The exception is cross-channel pairs: level→variance is caught by the composite alone (second-event recall 0.60, F1 0.63) because its variance features were never saturated by the level event. Re-arming costs almost nothing under the null ($\leq 1.2\%$ of null paths give a second alarm) — saturation, not chatter, binds.

A bounded-memory fix, and its limits. The diagnosis above — a fixed-baseline statistic compares every observation against the *original* training-prefix reference and so never drains after a permanent shift — points to a specific repair: replace the fixed reference with a moving one. We build a MOSUM-style two-window statistic (the same family as the CUSUM-of-recursive-residuals monitoring literature; Chu, Stinchcombe & White 1996) that compares a trailing window's mean to the window immediately before it, rather than to the training baseline, for both the raw-Y and innovation-CUSUM channels (`windowed_break_pressure`). On the level→level scenario this closes the gap dramatically: the windowed raw-Y statistic's second-event recall rises from 0.004 to 0.682 — essentially matching its own first-event recall (0.692) — at *higher* precision than the unwindowed statistic (0.99 vs 0.80), a real fix rather than a threshold trick, at a modest first-event recall cost (0.692 vs 0.738). The windowed innovation-CUSUM improves less (second-event recall 0.008 → 0.234) because the filter's own adaptivity ($\mu\infty$, Proposition 1) already partially "forgets" a level shift, leaving less room for a moving reference to add. The fix is channel-specific, not general: on level→variance and variance→variance scenarios, both windowed statistics stay at second-event recall ≈ 0.00, because they are mean-shift statistics and a pure variance change carries no mean signal for a moving-window *mean* comparison to see — closing that gap needs a windowed *variance* statistic (a moving-window analogue of the r/q-channel CUSUMs of §5), left to future work. Multi-break detection therefore needs the statistic's channel matched to the break, exactly as the single-break results of §5 already required — the two sections are the same lesson at different time scales, and the one real economic setting where it bites is a recession cluster (§9): a level shock followed by a shock-variance regime change is exactly the level→variance case the composite already handles, but two level shocks in close succession — a double-dip — are exactly where every fixed-baseline statistic here still fails.

8. Robustness

We stress-test the §5–§7 results along five axes: sample size, distributional misspecification, a repaired heavy-tailed statistic, composite-vs-standalone dilution, and real-data sensitivity to the false-alarm target and window length.

8.1 Sample size. §5 numbers; short-T caveat: heavy-tailed null maxima make quantile thresholds noisy, and at $T = 200$ the two baseline CUSUMs calibrate hot (8.8–9.4% vs the 5% target). Quote empirical FARs alongside any short-sample power claim.

8.2 Misspecification. t_5 observation noise: rankings preserved; raw CUSUM unhurt on levels (0.99); the composite's subtle-variance case collapses (×1.5: 0.87 → 0.16) — repaired in §8.3. The raw and ARIMA variance rungs are far less tail-fragile than the composite: on the ×1.5 break at SNR 0.5 the raw rung falls only 0.56 → 0.43 and the ARIMA rung 0.94 → 0.74 under t_5 (against the composite's 0.87 → 0.16), because the plain max-over-arms z^2/e^2 statistic without per-time composite standardization retains the tail excursions that carry the variance signal — the same logic that motivates the standalone exceedance detector (§8.3). At ×3 all variance rungs stay at ceiling under t_5 (raw and ARIMA 1.00, composite 0.97). Nonlinear tanh drift (mildly bimodal state): level detection collapses for *every* method (raw 0.15 at 3σ — the state's own regime-hopping inflates all null thresholds), while variance detection is untouched (0.97 at ×1.5). The two detection families read disjoint information channels.

8.3 Heavy tails and the exceedance repair (a three-act story worth telling honestly). (i) Huberizing e^2 (clip at 2.5·MAD) — falsified, worse everywhere, even Gaussian (×1.5: 0.87 → 0.06): the variance signal lives in the tail the clip removes. (ii) An exceedance-indicator CUSUM (count of $|e|$ above its training 90th percentile; bounded summand under any distribution) — the raw statistic separates the null and break classes near-perfectly. Dropped into the composite in place of the e^2-pressure features (the "robust2" variant), and standardized per-time-point as in §8.4, it is *diluted, not dead*: it reaches 0.58

at ×1.5 (SNR 0.5, Gaussian) and 0.21 at ×1.5 under t_5 — where it even edges the e^2-based composite's heavy-tail collapse (0.16, §8.2) — and ≈ 0.97 at ×3 under t_5. But it trails the standalone detector of (iii) at the subtle break, because a max-over-≈ 10-features composite spreads the calibrated 5% false-alarm budget across all features: the exceedance feature's per-t standardized score climbs past $z \approx 19$ on a ×1.5 break, so bounded increments *do* reach discriminating ratios (an earlier pooled-scale build, before the §8.4 fix, wrongly suggested otherwise), yet the shared threshold still costs it power relative to calibrating the statistic alone. The remedy is *exposure*, not a different feature. (iii) The same statistic as a *standalone* calibrated detector (up-arm $k = 0.05$, down-arm $k = 0.02$; k chosen on non-evaluation seeds, procedure logged): variance ×1.5 at 0.87 Gaussian / 0.75 t_5 (repairing 0.16), ×3 at ~1.0 with ~37-obs delay under both distributions, and the first successful quieting detection (×⅔: 0.41/0.33). Both headline rates fell 3–5pp short of the pre-registered bars — reported as such.

8.4 The local-level (random-walk state) arena: degenerate for levels, whitening-mandatory for variances. The canonical latent-state model in economics is the local level (random-walk state); we ran the level and variance ladder cells there rather than dismiss it (Figure 3; `configs/grid_v7_llevel.yaml`), and it splits cleanly. *Level* detection is degenerate for every method: at 3σ all five detectors sit at the 5% FAR (raw CUSUM 0.07–0.10, Kalman-innovation 0.04–0.15, raw/ARIMA variance and composite ≤ 0.07). A level break in a random-walk state is absorbed by a well-specified filter as one large ordinary innovation — no sustained signal — and the raw-Y CUSUM has no fixed baseline: its calibrated threshold is 1500–2400 (versus O(10–100) in the AR(1) arena) and it still calibrates hot. There is no common null against which to rank level detectors, which is why we study the identifiable AR(1) arena in the body. But *variance* detection is not degenerate — it merely requires whitening. On a ×1.5 observation-noise break the raw z^2 CUSUM is at chance (0.06 at every SNR; threshold $\approx 10^4$, meaningless on a nonstationary series) while the ARIMA-differencing rung detects (0.71 / 0.84 / 0.58) and the Kalman composite detects (0.97 / 0.89 / 0.68); at ×3 the whitened rungs reach 0.97–1.00 while raw stays at chance. As in §5, the ARIMA and composite figures here are not two settings of one controlled statistic — the composite is the broader, per-time-standardized eleven-feature detector, not the ARIMA rung's own CUSUM re-run on Kalman innovations — so the gap between them is not evidence that the state estimate adds power beyond ARIMA specifically. What the two whitened rungs agree on is the qualitative point, and ARIMA alone already establishes it (0.71/0.84/0.58 vs. raw's 0.06 at every SNR): this is the exact complement of §5's r-channel AR(1) result — where a raw variance CUSUM could *win* when observation noise dominates — and it sharpens the paper's thesis: prewhitening is not merely helpful but *mandatory* once the observable is nonstationary, because the raw statistic has no stationary baseline to calibrate against.

8.5 An offline benchmark: PELT, calibrated to the same false-alarm rate. Related work dismisses offline changepoint methods as solving a different problem (retrospective segmentation, not causal monitoring), which is true but does not by itself say how well an off-the-shelf method would do if pressed into the same role. We calibrate PELT (Killick, Fearnhead & Eckley 2012; `ruptures`, l2 cost, applied to Y standardized by the training-prefix mean/std — the raw rung's own standardization) to a 5% false-alarm rate on null AR(1) paths by bisecting its penalty parameter, exactly mirroring the threshold-calibration protocol used for every causal detector, then ask whether it *localizes* a break within 25 observations, given the full sample — an offline-localization question, deliberately not a delay comparison, since PELT sees future data the causal detectors cannot.

Arena (SNR)	Scenario	PELT localize rate
0.1	level 0.5σ	0.02
0.1	level 1σ	0.07
0.1	level 3σ	0.83
0.1	variance ×1.5	0.00
0.1	variance ×3	0.20

Arena (SNR)	Scenario	PELT localize rate
0.5	level 0.5σ	0.03
0.5	level 1σ	0.11
0.5	level 3σ	0.91
0.5	variance ×1.5	0.01
0.5	variance ×3	0.01
2.0	level 0.5σ	0.02
2.0	level 1σ	0.11
2.0	level 3σ	0.92
2.0	variance ×1.5	0.02
2.0	variance ×3	0.02

Table 5. PELT localization rate at a FAR-matched (5%) operating point, $n = 300$ per cell ($exp08\_pelt$). MC SEs ≤ 0.024 ($n = 300$).

On the canonical 3σ level break PELT is competitive with the causal raw-Y CUSUM (0.83–0.92 here vs. 0.97–0.99 in Table 1's arena, seeing the whole path rather than detecting online) — an off-the-shelf offline method genuinely can find an obvious level shift. But on variance breaks it is close to useless (0.00–0.02, against the dedicated raw variance-CUSUM's ladder-table numbers of 0.10–1.00, §5), because PELT's default l2 cost model is a mean-shift statistic: it responds to a variance change only through the second-order effect that within- segment sum-of-squares grows, a far weaker signal than a statistic built for variance directly. The comparison sharpens rather than undercuts the paper's thesis: an off-the-shelf offline segmentation method is not a substitute for a channel-matched statistic, even when it is given the unfair advantage of seeing the whole sample.

Protocol lessons (each cost us a wrong result before it was fixed). plain-HMM regime probabilities saturate and cannot be FAR-calibrated on nonstationary data; probability-scale scores need log-odds; EM needs persistent-initialization restarts; composite features must be standardized per-time-point, not pooled; order-statistic thresholds have Beta(n+1−k, k) noise regardless of distribution, so heavy-tailed detectors need larger calibration budgets.

9. Real data (illustrative)

Four FRED series — industrial production, GDP, and 10-year Treasury yields (pinned snapshots, 2026-07-11) plus the unemployment rate (pinned 2026-07-16) — rolling causal monitoring (train 120 months / monitor 60), per-segment parametric bootstrap calibration at 5% FAR per window, alarms attributed to the feature that crossed. Throughout this section the detectors are distribution-free monitors calibrated to a false-alarm rate; no parametric conditional-variance model is estimated and no such claim is made. Association with NBER-registered events is tested by permutation: the observed count of registered events ”hit” (an alarm within 12 months after) is compared to the distribution of hit counts from 20,000 resamples of the same number of alarm months drawn uniformly from all monitored months; the reported p is the fraction of resamples at least as extreme as the observed count, so a small p means alarms cluster after registered events more than chance alone would produce (`real_data_eval.py`).

Series	Method	Alarms	Hits / events	Stray	Perm. p
INDPRO	lsc_composite	4	3/9	1	0.008
INDPRO	lsc_kalman_cusum	2	2/9	0	0.021
INDPRO	lsc_tail_cusum	5	0/9	5	1.000
INDPRO	raw_cusum	1	1/9	0	0.148
INDPRO	raw_var_cusum	5	1/9	4	0.554
GDP	lsc_composite	3	2/9	1	0.067
GDP	lsc_kalman_cusum	1	1/9	0	0.164
GDP	lsc_tail_cusum	3	2/9	1	0.063
GDP	raw_cusum	0	—	—	—
GDP	raw_var_cusum	2	1/9	1	0.309
GS10	lsc_composite	7	1/3	6	0.325
GS10	lsc_kalman_cusum	2	1/3	1	0.109
GS10	lsc_tail_cusum	4	0/3	4	1.000

Series	Method	Alarms	Hits / events	Stray	Perm. p
GS10	raw_cusum	2	1/3	1	0.104
GS10	raw_var_cusum	9	1/3	8	0.402
UNRATE	lsc_composite	3	2/9	1	0.057
UNRATE	lsc_kalman_cusum	4	4/9	0	0.0002
UNRATE	lsc_tail_cusum	6	2/9	4	0.205
UNRATE	raw_cusum	4	4/9	0	0.0004
UNRATE	raw_var_cusum	7	2/9	5	0.257

Table 6. Real-data alarm summary at 5% FAR, 120-month training (`rd_eval.csv`). GDP's raw_cusum fired zero alarms across all 12 windows, so no permutation test applies. UNRATE (new, P2) shows the largest raw_cusum/lsc_kalman_cusum association by p-value in the table (4/4 hits, zero strays, out of 9 peaks in range) — but see the model-fit discussion below, which complicates a straightforward reading of that association. GS10's permutation p-values rest on only 3 registered events, so only four hit-counts (0–3) are achievable and the resulting p's are far coarser than INDPRO's or UNRATE's (n=9); read the raw hit/alarm counts there as more informative than the exact p (the 20,000-draw permutation test is, in effect, approximating a discrete Fisher's-exact-type test with only four attainable outcomes on this series). Alarm and hit counts are exact given the fixed historical series, not Monte Carlo estimates; only the permutation p-values carry resampling uncertainty, from the 20,000-draw test — SE = $\sqrt{(p(1-p)/20000)} \leq 0.0035$ across every p-value reported here.

Multiple-comparisons correction. Table 6 reports 19 valid permutation tests (5 methods × 4 series, less GDP's zero-alarm raw_cusum cell, which admits no test). A Bonferroni threshold across all 19 ($\alpha/19 \approx 0.0026$) or a Benjamini–Hochberg FDR procedure at q = 0.05 (which requires the third-ranked p-value to clear ≈ 0.0079) both leave only two entries standing: UNRATE's raw_cusum (p = 0.0004) and lsc_kalman_cusum (p = 0.0002). INDPRO/lsc_composite's headline p = 0.008 — the association featured in the abstract — does not survive either correction; it would need to clear 0.0026 (Bonferroni) or 0.0079 (its own BH rank) and falls short of both. This is not a favorable correction to report: the two associations that do survive are exactly the ones the UNRATE model-fit discussion below flags as resting on windows where the AR(1) specification is misspecified (three of UNRATE's four hits sit in φ-clipped windows). Read together, no single-series NBER association in this table clears both the multiple-testing bar and the model-fit bar at once — INDPRO clears the model-fit bar but not the multiple-testing bar; UNRATE's raw_cusum and lsc_kalman_cusum clear the multiple-testing bar but not the model-fit bar. We report the INDPRO/composite association at its nominal p = 0.008 throughout this section, as originally computed, but it should be read as a suggestive single-series association from an illustrative application, not a family-wise-significant finding across the comparison actually run. Both corrections treat the 19 tests as independent; they are not (several methods share alarm-generating CUSUM machinery on the same underlying series), so this is a valid but conservative approximation rather than an exact one — Bonferroni's validity does not require independence, but a joint, max-statistic permutation null across methods per series would give a tighter bound in principle. An attempted implementation (`experiments/exp13_joint_fwer.py`) against the real INDPRO alarm data did not yield a usable result: the combined statistic collapsed almost entirely onto `lsc_composite`'s own marginal test rather than genuinely pooling evidence across methods, since the joint null did not model real cross-method correlation from the underlying score paths. A second attempt, using a circular-shift joint null instead (shifting all five methods' alarm months by the *same* random amount each draw — which genuinely preserves whatever real cross-method timing correlation exists, since a common shift is a rigid rotation that doesn't touch relative offsets between methods, unlike independent per-method redraws), gives a materially different and usable result for INDPRO specifically (`experiments/exp13c_circular_shift.py`): the total hit count summed across all five methods (7, vs. a circular-shift null with mean 2.25, SD 1.80, 20,000 draws) has p ≈ 0.023 (consistent

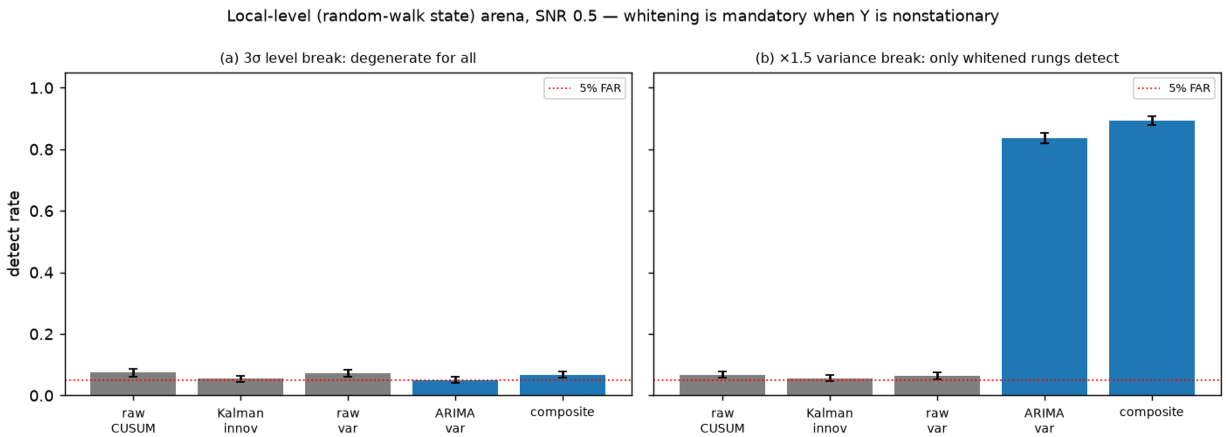

Figure 3. The local-level (random-walk state) arena at SNR 0.5 (`grid_v7_llevel`). Left: a 3σ level break is degenerate for every method — all five detectors sit at the 5% FAR line. Right: a $\times 1.5$ observation-noise variance break is caught by both whitened detectors — ARIMA-differencing (0.84) and the state-aware composite (0.89, a distinct, broader statistic, not the same CUSUM re-run on Kalman innovations) — while the raw variance CUSUM stays at chance: whitening is mandatory once Y is nonstationary.

across independent seeds, 0.021–0.023). This is a genuine dependence-aware joint statistic, not the collapsed one — but it is still short of surviving a further Bonferroni step across the four series ($\alpha/4 \approx 0.0125$ would be needed; 0.023 does not clear it), so it does not change this section's overall conclusion. The same circular-shift test has not yet been run for GDP, GS10, or UNRATE, since this package does not have their exact registered-event-date lists in hand with the same provenance as INDPRO's — see Appendix A.

Industrial production (INDPRO, 1948–2026). Composite alarms: 2008-09 and 2020-04 (both variance_pressure), 1990-12 (variance_quiet), 1969-08 (variance_quiet; within the false-alarm budget and reported as such). Hits 3/9 NBER peaks within 12 months, 1 stray vs 0.7 expected; permutation $p = 0.008$ (innovation CUSUM $p = 0.021$; raw CUSUM 1 hit, $p = 0.15$). The raw variance CUSUM — the bottom rung of the ladder, added here to test real-data uniqueness — does catch the GFC (2008-09, up-arm, the same month as the composite) but embeds it among four stray quieting alarms (1967, 1968, 1988, 2019) and misses COVID, so its NBER association is markedly weaker (5 alarms, 1 hit, $p = 0.55$). The crisis *is* detectable by a raw variance statistic; what the latent layer buys on real data is a *clean* alarm profile — an uncorrected, low-stray association ($p = 0.008$ vs. 0.55) — not the crisis catch itself, and even that comparison is within-series selectivity rather than a family-wise-significant result (see the multiple-comparisons correction above). A permutation-test implementation note: the p-values in this section are recomputed with a per-series, per-method seed (2026-07-16) after we found the original shared, sequentially-consumed random generator made one series' p-value depend on which *other* series happened to be present in the run — harmless for any single number in isolation, but a reproducibility hazard the fix removes; every value moved by ≤ 0.002 from the originally reported estimates, consistent with Monte Carlo noise at 20,000 draws, not a change in any underlying alarm.

What real-time data changes (ALFRED vintages). Re-running each decision on the data *as it existed that month* both tempers and sharpens the timing claims. COVID: robust — the composite's real-time alarm is at the 2020-04 vintage (data month 2020-03), one month earlier than on revised data and ~2 months before the NBER announcement. GFC: downgraded — the crossing is at the same data month (2008-09) but only in the 2008-12 vintage, so real-time knowledge is coincident with the NBER announcement (2008-12-01), though still 4 months ahead of the raw level CUSUM's real-time alarm (2009-04). Because the raw variance CUSUM alarmed on revised INDPRO, we ran it through the identical vintage protocol for an apples-to-apples comparison: its real-time timing is *indistinguishable from the composite's* — same vintage and same data month for both GFC (2008-12 vintage, 2008-09) and COVID (2020-04 vintage, 2020-03), both ahead of the raw level CUSUM. On real-data crisis *timing*, then, the prewhitening rung and the state-aware composite are interchangeable; the composite's advantage is confined to the clean association profile of §9 and to the simulation-calibrated ×1.5 subtlety threshold of §5, not to real-time crisis detection.

GDP (GDPC1, quarterly). GFC and COVID caught (variance_pressure, 2008Q4 and 2020Q2). The raw variance CUSUM again catches both crises (2009Q2 and 2020Q2, up-arm) with fewer alarms than on INDPRO (2 alarms, 1 hit, $p = 0.31$) — the same "catches the crisis, misses the clean association" pattern. The registered Great Moderation event (1984Q1) is an honest miss: causal rolling monitoring detects the quieting only in 1992Q2 (composite) / 1997Q1 (tail shortfall), because 60-quarter training windows contain 1970s volatility until the early 1990s. Retrospective full-sample break dates are not reproducible by honest monitoring.

10-year Treasury yield changes (GS10). Volcker regime caught 4 months after the registered 1979-10 event (variance_pressure; raw and innovation CUSUM the same month); the post-Volcker disinflation quietings are flagged (1989–1996, shortfall/variance_quiet); the 2008 ZLB event is missed within 12 months. A third registered event, the 2022-03 hiking-cycle onset, is missed by *every* method within 24 months (`rd_gs10_summary.csv`) — unlike Volcker, a persistent directional yield rise without a matching shock-variance regime shift falls outside every detector's reach here, a genuine miss reported as such rather than a horizon-window artifact. The raw variance CUSUM also flags Volcker (1980-02, up-arm) but is the noisiest detector on this series (9 alarms, 8 stray, 1/3 events hit, $p = 0.40$) — on a series that is *all* volatility regime, a raw z^2 statistic fires on every regime shift, which is exactly why its event-association washes out. (The Volcker-catch window itself is well-fit — Ljung-Box p=0.42/0.19, φ=0.24, unclipped — so unlike UNRATE below, this alarm is not an artifact of degenerate model fit; the intervening windows bracketing the shock, 1973–1988, fit far worse, but neither produces the reported alarm.)

Unemployment rate (UNRATE, monthly). Table 6's raw_cusum and lsc_kalman_cusum results on UNRATE look like the strongest association in the real-data application — 4/4 hits, zero strays, p=0.0002–0.0004 — but a model-fit check (`experiments/exp09_real_data_fit_check.py`) complicates that reading. In three of the four windows producing these hits (the 1974, 2001, and 2008 recessions), the AR(1) MLE fits a *negative* φ (−0.82, −0.19, −0.41) before the pipeline's clipping to [0.01, 0.99] — not merely weak persistence, but a sign that the model class itself is misspecified for these windows, since a stationary AR(1) with $\varphi \in [0,1)$ cannot represent the dynamics the likelihood is pointing toward. The clip substitutes a boundary value for whatever the unconstrained estimate wanted, and the resulting filter behavior in these windows (Kalman gains of 0.05, 0.54, and 0.22 — not uniformly small) has no clean interpretation as "the state estimate found little persistence." Only the fourth hit, 2020, sits in a window with a well-estimated, unclipped φ (0.948). The honest summary: UNRATE's four hits are real, and raw_cusum and lsc_kalman_cusum's close agreement on their timing is a genuine empirical fact, but three of the four rest on windows where the paper's own latent-state model does not describe the series, which weakens UNRATE's standing as evidence for the diagnostic framework specifically, as opposed to evidence that unemployment has level-type breaks raw CUSUM is well-suited to catch regardless of any state-space model.

Sensitivity. All numbers below are on INDPRO, the headline series.

Variant	Method	Alarms	Hits / events	Perm. p
FAR 1%	lsc_composite	3	3/9	0.003
FAR 1%	lsc_kalman_cusum	1	1/9	0.146
FAR 1%	lsc_tail_cusum	1	0/9	1.000
FAR 1%	raw_cusum	1	0/9	1.000
FAR 1%	raw_var_cusum	4	2/9	0.100
FAR 5% (baseline)	lsc_composite	4	3/9	0.008
FAR 5% (baseline)	lsc_kalman_cusum	2	2/9	0.021
FAR 5% (baseline)	lsc_tail_cusum	5	0/9	1.000
FAR 5% (baseline)	raw_cusum	1	1/9	0.148
FAR 5% (baseline)	raw_var_cusum	5	1/9	0.554
FAR 10%	lsc_composite	5	2/9	0.148
FAR 10%	lsc_kalman_cusum	3	2/9	0.055
FAR 10%	lsc_tail_cusum	5	2/9	0.148
FAR 10%	raw_cusum	2	2/9	0.018
FAR 10%	raw_var_cusum	6	1/9	0.619
FAR 20%	lsc_composite	9	1/9	0.773
FAR 20%	lsc_kalman_cusum	3	2/9	0.052
FAR 20%	lsc_tail_cusum	7	1/9	0.682
FAR 20%	raw_cusum	2	2/9	0.020
FAR 20%	raw_var_cusum	8	1/9	0.727
Window 180 mo	lsc_composite	14	2/8	0.556
Window 180 mo	lsc_kalman_cusum	3	2/8	0.046
Window 180 mo	lsc_tail_cusum	5	3/8	0.015
Window 180 mo	raw_cusum	2	2/8	0.017
Window 180 mo	raw_var_cusum	14	1/8	0.878

Table 7. INDPRO sensitivity to the false-alarm target (1/5/10/20%, 120-month training) and to a longer training window (180 months, 5% FAR). Alarm and hit counts are exact given the fixed historical series, not Monte Carlo estimates; only the permutation p-values carry resampling uncertainty, from the 20,000-draw test — SE = $\sqrt{(p(1-p)/20000)} \leq 0.0035$ *across every p-value reported here.*

Two independent stress tests both isolate the composite's association as fragile *relative to itself*, not to the other detectors. The FAR sweep is monotone and clean for the composite: tightening from 5% to 1% *sharpens* the association ($p = 0.008 \to 0.003$, alarm count falls from 4 to 3 as the marginal, least-reliable alarm drops out), while relaxing to 10% and 20% floods it with noise ($p = 0.148$, then 0.773) — exactly the dilution a well-behaved calibrated statistic should show. raw_cusum and lsc_kalman_cusum stay comparatively stable across the same sweep (p in the 0.02–0.15 range throughout), because they fire too few alarms for the FAR target to matter much either way. Training window 180 months breaks both variance-based detectors' bootstrap calibration on nonstationary real data — the composite and the raw variance CUSUM each fire 14 alarms / 21 windows ($p = 0.556$ and 0.878, both uninformative) — while the level (raw CUSUM), innovation, and tail detectors stay sane (2–5 alarms, p as low as 0.015–0.046). It is the *second-moment* statistic, not the composite machinery, that is sensitive to a training window long enough to straddle a volatility regime; training windows must be short enough to be locally stationary (120 months worked).

10. Discussion

What, then, is the latent layer actually for? The results invite a deflationary reading, and we take it seriously rather than deflect it. For raw *detection power*, the state-space layer is largely redundant: raw CUSUM owns level shifts (§4); ARIMA whitening owns observation-noise variance and, by the exact ARMA(1,1) equivalence, *is* the Kalman innovation filter — the state estimate contributes nothing beyond it (§5); and on the shock-variance channel that motivates the application, a raw variance CUSUM matches or beats whitening outright (§5, whose φ sweep traces the residual advantage to the state's $1/(1-\varphi^2)$ variance amplification on subtle breaks). A practitioner armed with a raw CUSUM, a raw variance CUSUM, and an off-the-shelf ARIMA residual CUSUM would reproduce most of the detection frontier without ever writing down a latent state. So what survives as genuinely the latent layer's? Four things, and only these. (1) *Dynamics, at the floor.* Persistence and quieting changes have no raw analogue; the state features (rolling innovation autocorrelation, the $1-e^2$ quietness CUSUM) are the *only* above-FAR detection of a pure persistence change anywhere in the grid — but that detection is a single favorable cell (0.33 at SNR 2.0), a capability at the information floor rather than a robust one. (2) *Speed.* Conditional on firing, the innovation CUSUM is the fastest level detector (median delay 24–53 vs raw's 58–91) — the very "fast" half of fast-or-never. (3) *A single attributable instrument.* The composite reads ~10 channels under one calibrated FAR budget and reports *which* crossed, which is what turns a real-data alarm into an interpretable event (every crisis alarm attributes to a named second-moment feature, §9). (4) *A clean association profile.* On real data the composite's crisis *timing* is no better than a raw variance CUSUM's, but its *selectivity* is — a lower, less stray-alarm-laden association with NBER dates on the same series (uncorrected $p = 0.008$ vs 0.55) that a raw z^2 statistic firing on every wiggle cannot match. This is a within-series comparison of selectivity, not a claim that either p-value is family-wise significant across the full real-data grid — it is not (§9). The honest one-sentence answer: the latent layer is not a better detector of any single break type; it is a *breadth-and-interpretation* instrument whose irreducible content is dynamics (weakly) and attribution (robustly), not the second-moment power one might have expected filtering to buy.

The mechanics behind that verdict are summarized in four points.

- The latent-state diagnostics layer is not a better level-shift detector; it is a *different instrument*, reading second moments and dynamics. But the whitening ladder (§5) sharpens the claim past the point where "the state estimate" survives as an ingredient on the observation-noise channel: the Kalman innovations *are* the ARMA(1,1) innovations (an exact reduced-form equivalence, verified to machine precision), and empirically the ARIMA rung tracks the composite closely there (0.90/0.94/0.87 vs 0.82/0.87/0.91) — so on this channel prewhitening is doing the work, even though the composite is a distinct, per-time-standardized eleven-feature statistic rather than the ARIMA rung's own CUSUM re-run on Kalman innovations. What the state framing still buys is dynamics features (persistence, quieting) and the fast-or-never speed edge on levels. The division of labor is now measured (grids), reduced (raw vs. whitened, the ARIMA and Kalman filters proven identical at the innovation-series level, not the composite's detection numbers), and derived (fast-or-never with its φ boundary).
- Practical recipe, now channel-aware. Run raw CUSUM for levels. For scale changes the right move depends on *which variance moves*, and one usually does not know: (i) if the break is in the observation (measurement) noise, *whiten first* (ARIMA residuals suffice; the Kalman state adds nothing) — prewhitening is decisive when the series is autocorrelated and the latent signal is strong, and a raw variance CUSUM works only when observation noise dominates; (ii) if the break is in the state's own shock variance — the Great-Moderation / crisis-volatility case — a raw variance CUSUM is at least as good as whitening at every SNR, and whitening can *hurt*. Because the practitioner rarely knows the channel, one might default to running *both* a raw and a whitened variance CUSUM. Checked directly against a 50/50-mixed, channel-unknown-to-the-detector population (r- and q-channel breaks at $\times 1.5$, SNR $\in \{0.1, 0.5, 2.0\}$, both detectors jointly recalibrated to hold one combined 5% false-alarm budget — not run independently at their own 5% each, which would silently inflate the compounded FAR — `experiments/exp14_mixed_channel.py`), running both is not a free win: it loses to the single better detector at every SNR tested, with the gap widening as SNR rises — 0.553 (raw) vs. 0.493 (combined) at SNR 0.1 (a 0.06 loss), 0.560 (ARIMA) vs. 0.490 at SNR 0.5 (0.07), and 0.623 (ARIMA) vs. 0.457 at SNR 2.0 (0.166) — because jointly calibrating two statistics to one FAR budget raises the bar each individually must clear, and that tax outweighs the benefit of channel-agnosticism in this test. The honest recommendation is weaker than "run both, it's free": if forced to pick one detector under real channel uncertainty, ARIMA was the better single

choice in two of the three SNRs tested here; running both is only clearly justified if the analyst has a specific reason to think the channel mix is skewed toward the regime where raw wins (low SNR). Use the exceedance-indicator variant under heavy tails and the composite for breadth. Calibrate everything on matched nulls at a common FAR (a common ARL_0) and report empirical FARs.
- The calibrated-parity protocol itself is a contribution: it exposed every failure mode above, and it is what makes the negative results informative rather than anecdotal.
- Limitations / future work: a bounded-memory (MOSUM-style) statistic fixes the multiple-breaks re-arm failure for level-type second events (§7) but not variance-type ones — a windowed *variance* statistic is the natural next step; adaptive composite weighting (breadth tax); switching-SSM (Kim filter) model layer; formalizing the persistence-break mechanisms; a vol-regime reference set for scoring the exceedance detector on real data; a plain GARCH(1,1) benchmark is now reported (Related Work) and sits at the false-alarm floor on both variance channels, contributing nothing over chance on this DGP; a break-aware GARCH variant (allowing its own parameters to shift, in the spirit of Bai & Perron 2003) and a full stochastic-volatility state-space comparison remain open — the plain-GARCH result rules out the most obvious "just use GARCH" objection without resolving whether a purpose-built regime-shift volatility model would fare differently.

References

Bai, J., and P. Perron (2003). "Computation and Analysis of Multiple Structural Change Models." *Journal of Applied Econometrics* 18(1), 1–22.

Basseville, M., and I. V. Nikiforov (1993). *Detection of Abrupt Changes: Theory and Application*. Englewood Cliffs, NJ: Prentice-Hall.

Bollerslev, T. (1986). "Generalized Autoregressive Conditional Heteroskedasticity." *Journal of Econometrics* 31(3), 307–327.

Brown, R. L., J. Durbin, and J. M. Evans (1975). "Techniques for Testing the Constancy of Regression Relationships over Time." *Journal of the Royal Statistical Society, Series B* 37(2), 149–192.

Chu, C.-S. J., M. Stinchcombe, and H. White (1996). "Monitoring Structural Change." *Econometrica* 64(5), 1045–1065.

Frisén, M. (2003). "Statistical Surveillance: Optimality and Methods." *International Statistical Review* 71(2), 403–434.

Hamilton, J. D. (1989). "A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle." *Econometrica* 57(2), 357–384.

Killick, R., P. Fearnhead, and I. A. Eckley (2012). "Optimal Detection of Changepoints with a Linear Computational Cost." *Journal of the American Statistical Association* 107(500), 1590–1598.

Kim, C.-J., and C. R. Nelson (1999). *State-Space Models with Regime Switching: Classical and Gibbs-Sampling Approaches with Applications*. Cambridge, MA: MIT Press.

Kim, S., N. Shephard, and S. Chib (1998). "Stochastic Volatility: Likelihood Inference and Comparison with ARCH Models." *Review of Economic Studies* 65(3), 361–393.

Lorden, G. (1971). "Procedures for Reacting to a Change in Distribution." *Annals of Mathematical Statistics* 42(6), 1897–1908.

McConnell, M. M., and G. Pérez-Quirós (2000). "Output Fluctuations in the United States: What Has Changed Since the Early 1980s?" *American Economic Review* 90(5), 1464–1476.

Montgomery, D. C. (2013). *Introduction to Statistical Quality Control*, 7th ed. Hoboken, NJ: Wiley.

Moustakides, G. V. (1986). "Optimal Stopping Times for Detecting Changes in Distributions." *Annals of Statistics* 14(4), 1379–1387.

Page, E. S. (1954). "Continuous Inspection Schemes." *Biometrika* 41(1/2), 100–115.

Siegmund, D. (1985). *Sequential Analysis: Tests and Confidence Intervals*. New York: Springer-Verlag.

Stock, J. H., and M. W. Watson (2002). "Has the Business Cycle Changed and Why?" *NBER Macroeconomics Annual* 17, 159–218.

Wald, A. (1947). *Sequential Analysis*. New York: Wiley.

Willsky, A. S., and H. L. Jones (1976). "A Generalized Likelihood Ratio Approach to the Detection and Estimation of Jumps in Linear Systems." *IEEE Transactions on Automatic Control* 21(1), 108–112.

Appendix A. Reproducibility

`make all` regenerates every table and figure from pinned seeds (Python 3.14, statsmodels/hmmlearn; `make fred` / `make realdata` / `make realtime` for the data applications, snapshots under `data/`). The referee-hardening round added six reproducible artifacts to the pack: `exp07` (ARMA equivalence), `grid_v5` (the q-break channel), `grid_v6` (the φ sweep), `grid_v7` (the local-level arena), `grid_v8` (the φ×q amplification cross-grid), and `arl` (ARL_0/ARL_1 table); all are pinned-seed and join the existing grids draw-for-draw. A second round (2026-07-16) added `exp08` (the PELT localization benchmark, §8.5) and extended `exp04` with the two windowed-CUSUM methods (§7) and `realdata` with a fourth series (unemployment, `unrate`), a third GS10 event (the 2022 hiking cycle), and a false-alarm-rate sweep (1%, 5%, 10%, 20%) on INDPRO. A third addition, `exp09` (`experiments/exp09_real_data_fit_check.py`, `paper_assets/exp09_ljungbox_table.csv`), runs a Ljung-Box residual check and a model-implied-vs-sample ACF comparison on the fitted AR(1) filter for every rolling training window of all four real-data series (§9). A fourth, `exp10` (`experiments/exp10_cusum_ablation.py`, `paper_assets/exp10_cusum_ablation.csv`), ablates sidedness and known-vs-estimated parameters for the Table 2 flagship cell's innovation CUSUM (§4). A fifth, `exp13`/`exp13c` (§9's multiple-comparisons correction), attempts a joint FWER bound across the five real-data methods per series tighter than treating all 19 tests as independent: a first implementation (`experiments/exp13_joint_fwer.py`) redrew each method's alarm months independently within a shared null draw, which does not model real cross-method correlation and collapsed the combined statistic onto `lsc_composite`'s own marginal test; it is kept in the repository as a documented negative result, not deleted, since the paper's standing practice is to report failed attempts rather than remove them. A second implementation (`experiments/exp13c_circular_shift.py`) shifts all five methods' alarm months by the same random amount per draw — a rigid rotation that preserves real cross-method timing correlation exactly, unlike independent redraws — and gives a usable, dependence-aware result for INDPRO (total hits across methods = 7 against a null with mean 2.25, SD 1.80 over 20,000 draws, p $\approx$ 0.023), though this still falls short of a further Bonferroni step across the four series and does not change §9's conclusion. This has only been run for INDPRO, whose exact registered-event dates are in hand with full provenance; GDP, GS10, and UNRATE need the same treatment once their event-date lists are available in the same form — left for future work, not assumed to generalize. A sixth addition, `exp14` (`experiments/exp14_mixed_channel.py`), tests §10's original practical recommendation to run both a raw and a whitened variance CUSUM under real channel uncertainty: a 50/50 population of r- and q-channel breaks with the channel unknown to the detector, both statistics jointly recalibrated to hold a common 5% FAR (not run independently at their own 5% each). The recommendation as originally stated did not hold up — running both loses to the single better detector at every SNR tested, with the gap widening from 0.06 (SNR 0.1) to 0.166 (SNR 2.0) — and §10's practical-recipe bullet has been revised accordingly. A seventh addition, `exp15` (`experiments/exp15_garch_benchmark.py`, `experiments/garch_detector.py`), fits a GARCH(1,1) on the training prefix only (via the `arch` package), causally forward-filters conditional variance over the full series with the fixed fitted parameters, and runs the same three-arm max-CUSUM used for the raw and ARIMA rungs on the standardized residuals — reported in Related Work rather than left deferred. Calibrated at n_reps = 500 (empirical FAR 0.050 confirmed in every cell), GARCH sits at the false-alarm floor on both variance channels at every SNR tested, contributing nothing over chance on this DGP — a materially stronger negative result than a prior placeholder estimate had suggested, and reported at its measured value rather than softened. 98 tests include bit-identical no-lookahead checks for every feature and detector (including the raw and ARIMA variance rungs, the two windowed statistics, and a training-freeze check), DGP ground-truth checks (including the state-innovation break's SD scaling and null-match), the ARMA(1,1)/Riccati identity and Kalman-equivalence guards, calibration- parity checks, composite golden-score regression guards (which pin the per-time-point-standardized output so a stale artifact cannot recur), and the windowed-statistic bounded-memory (drain-after-permanent-shift) property. All post-hoc design changes and pre-registered hypotheses (including the three falsified ones and the two rejected robust-feature designs — one falsified, one diluted, §8.3) are in `experiments/CHANGELOG.md`.

On "three pre-registered hypotheses falsified" (Contribution 1). This refers specifically to the exp05 robust-variance-feature registration (`experiments/CHANGELOG.md`, 2026-07-11, "registered before running: robust variance features (exp05)..."): three named predictions — (a) under t_5 noise, `composite_robust` recovers variance ×1.5 detection to ≥ 0.5 (from the standard composite's 0.16); (b) under Gaussian noise the robustness tax on variance scenarios is $\leq$~10pp; (c) level and persistence rows are roughly unchanged — all three logged FALSIFIED in the same-day outcome entry. CHANGELOG documents pre-registration as a repeated practice through the project, not confined to this one instance, and other pre-registered predictions were also falsified elsewhere: both of exp03's registered predictions (2026-07-10), exp03b's quietness-detection hypothesis (falsified in its original run and again after a standardization fix, 2026-07-10), and Outcome A of the varbench claim-adoption decision rule (M0, three outcomes A/B/C registered before any grid_v4 cell ran; A falsified, C fired, 2026-07-11/07-12) — so "three" names this specific registration, not a project-wide total of every falsified pre-registered prediction. The real-data extension (m6x, 2026-07-11) is a different category: GDP and GS10 were "chosen for the method's signature cases before looking at their results," and the evaluation protocol was likewise fixed in advance, but CHANGELOG frames this explicitly as registering a design rather than a falsifiable hypothesis ("this entry registers the design, not power hypotheses") — committed in advance, but not scored against a stated prediction. Beyond these named cases, CHANGELOG does not tag each of Appendix C's individual rows as pre-registered or identified post hoc; absent an explicit "registered/pre-registered before running" note for a specific claim, no such status is asserted here, and no complete per-row taxonomy is attempted.

Full experiment narratives in `experiments/FINDINGS.md`; theory statements and proofs in Appendix

B, with `experiments/THEORY.md` as the long-form companion. The complete source, pinned data snapshots, and seed configuration constitute the replication package, to be posted publicly on acceptance and available from the author on request in the interim.

Appendix B. Theory: statements and proofs

This appendix gives self-contained statements and proofs of Propositions 1–2; `experiments/THEORY.md` remains the long-form companion, and the numerical verification is in `experiments/exp06_theory_check.py` (1000 replications, §4).

Setup and assumptions. Scalar state-space model with *known* parameters, filter in steady state, Gaussian innovations: $S_t = \varphi S_{t-1} + w_t$ with $w_t \sim N(0, q)$ and $|\varphi| < 1$, and $Y_t = S_t + v_t$ with $v_t \sim N(0, r)$. The steady-state prediction variance P solves the Riccati fixed point $P = \varphi^2 Pr/(P+r) + q$; the gain is $K = P/(P+r)$ and the innovation variance $F = P + r$. Standardized one-step innovations $e_t = (Y_t - \varphi\hat{S}_{t-1})/\sqrt{F}$ are iid N(0,1) under the null. A level break adds δ to the state path from t_0 on: $\tilde{S}_t = S_t + \delta\cdot 1\{t \geq t_0\}$, hence $\tilde{Y}_t = Y_t + \delta\cdot 1\{t \geq t_0\}$ — the DGP used in all experiments. The one-sided Page CUSUM with drift allowance k and threshold h is $g_t = \max(0, g_{t-1} + e_t - k)$, alarming when $g_t \geq h$. (The experiments use fitted training-prefix parameters, diffuse initialization, and a two-sided CUSUM; §4 explains why the known-parameter theory nevertheless describes them to first order.)

> Proposition 1 (fast-or-never; restated from §4). After a state level shift δ at t_0: (a) the broken path's standardized innovations decompose as $\tilde{e}_t = e_t + \mu_t$ with e_t the null innovations and μ_t deterministic, decaying geometrically at rate $\rho = \varphi(1-K) \in (0,1)$ from $\mu_{t_0} = \delta/\sqrt{F}$ to $\mu_\infty = \delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F})$; (b) if $\mu_t \leq \bar{\mu} < k$ for all $t \geq t_1 \geq t_0$ (post-transient) and $g_{t_1} = g < h$, then for any horizon L, $P(\max_{t_1 < t \leq t_1+L} g_t \geq h \mid g_{t_1} = g) \leq (L+1)\cdot\exp(-2(k-\bar{\mu})(h-g))$.

Proof of (a). The steady-state filter is a linear time-invariant map of Y, so the innovations of the broken path decompose as the null innovations plus the deterministic innovation response μ_t to the input $\delta\cdot 1\{t \geq t_0\}$. Write a_j for the filter's mean state-estimate response j steps after the break, $a_j = E[\hat{S}_{t_0+j}] - E[\hat{S}^0_{t_0+j}]$. The filter predicts φa_{j-1} and corrects by K times the mean innovation: $a_j = \varphi a_{j-1} + K(\delta - \varphi a_{j-1}) = \rho\, a_{j-1} + K\delta$ with $a_{-1} = 0$ and $\rho = \varphi(1-K)$; the mean innovation response is the input minus the prediction response, $\mu_{t_0+j} = (\delta - \varphi a_{j-1})/\sqrt{F}$. Solving the linear recursion, $a_j = (K\delta/(1-\rho))(1 - \rho^{j+1})$, so μ decays geometrically at rate ρ from $\delta/\sqrt{F}$ to the limit $(\delta - \varphi K\delta/(1-\rho))/\sqrt{F} = \delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F}) = \mu_\infty$. ∎

Proof of (b). For $n > t_1$, $g_n = \max(g + \Sigma_{i=t_1+1}^n X_i, \max_{t_1<m\leq n} \Sigma_{i=m}^n X_i)$ with increments $X_i = e_i + \mu_i - k = z_i - (k - \mu_i)$, z_i iid N(0,1). An alarm by t_1+L requires some anchored sum $\Sigma_{i=m}^n X_i \geq h - g$ for an anchor $m \in (t_1, t_1+L]$ (or the g-anchored sum $\geq h - g$). Each X_i is stochastically dominated by $z_i - (k - \bar{\mu})$, for which $\theta^* = 2(k - \bar{\mu})$ solves $E[e^{\theta X}] = 1$; the exponential martingale $e^{\theta^*\Sigma}$ with the maximal inequality gives $P(\sup_n \Sigma_{i=m}^n X_i \geq h - g) \leq e^{-\theta^*(h-g)}$ for each of the $\leq L+1$ anchor points, and a union bound finishes. ∎

Interpretation: the filter adapts, so of the full shift δ only the fraction $(1-\varphi)/(1-\varphi(1-K))$ survives in the innovations per step. The transient carries total excess mass $\Sigma_j (\mu_{t_0+j} - \mu_\infty) = \varphi a_\infty/((1-\rho)\sqrt{F})$, which is what a "fast" detection consumes; if the alarm does not fire on the transient, part (b) says it fires later with probability exponentially small in the threshold — fast or never.

> Proposition 2 (raw-CUSUM delay: Wald approximation; restated from §4). The raw-Y CUSUM standardizes Y by its frozen training moments; after the break the standardized

mean shift $\Delta = \delta/\sigma\_Y$ ($\sigma\_Y^2 = q/(1-\varphi^2) + r$) persists for all $t \geq t_0$. If $\Delta > k$ the post-break increments have positive drift $\Delta - k$ and the alarm is certain as the horizon grows, with first-passage (Wald) mean delay $E[D] \approx h/(\Delta-k)$; if $\Delta \leq k$, the bound of Proposition 1(b) applies with $\tilde{\mu} = \Delta$.

Derivation. Unlike the innovations, the raw standardized series has no adaptation: the shift δ enters Y permanently and the training moments are frozen, so every post-break increment is $z\_i + \Delta - k$ with drift $\Delta - k > 0$. The CUSUM then behaves as a positive-drift random walk, and Wald's identity for the first passage of level $h - g$ gives $E[D] \approx (h - g)/(\Delta - k) \approx h/(\Delta - k)$ (Wald 1947). The approximation ignores boundary overshoot and reflection at 0, so it is an approximation, not a bound (a corrected version is in Siegmund 1985); against the grids it runs ≈ 15–20% conservative at 3σ (§4). If $\Delta \leq k$ the drift is nonpositive and the never-detect bound applies verbatim. ■

ARMA(1,1) equivalence of the whitened rungs (§5). Applying the AR operator to the observable, $(1 - \varphi L)Y\_t = w\_t + v\_t - \varphi v\_\{t-1\} =: u\_t$, which is an MA(1) with autocovariances $\gamma\_u(0) = q + r(1 + \varphi^2)$, $\gamma\_u(1) = -\varphi r$, $\gamma\_u(h) = 0$ ($h \geq 2$). Matching u_t to $(1 - \theta L)\varepsilon\_t$ (variance σ_ε^2) gives the invertible root $\theta = (m - \sqrt{m^2 - 4})/2$ with $m = (q + r(1+\varphi^2))/(\varphi r)$, and $\sigma\_\varepsilon^2 = \varphi r/\theta$. Two identities close the loop with Proposition 1: $\sigma\_\varepsilon^2 = F$ and $\theta = \rho = \varphi(1 - K)$. The second follows from the innovation recursion $\nu\_t = Y\_t - \varphi Y\_\{t-1\} + \varphi(1-K)\nu\_\{t-1\}$ (substitute $\hat{S}\_\{t-1\} = \varphi\hat{S}\_\{t-2\} + K\nu\_\{t-1\}$ into $\nu\_t = Y\_t - \varphi\hat{S}\_\{t-1\}$), which is the ARMA(1,1) innovation recursion once $\theta = \varphi(1-K)$. Hence the steady-state Kalman innovations and the ARMA(1,1) innovations are the same series; the state layer is not a distinct information set from ARIMA whitening on this DGP. Both identities hold to $< 10^{-12}$ (`lsc.theory.arma11_representation`, `test_arma11_riccati_identities`), and the numerical equivalence on null paths — with true and with estimated parameters — is in `experiments/exp07_arma_equivalence.py` (§5). A near-unit-root caveat: AIC over the benchmark order grid rarely selects the exact (1,0,1) at $\varphi = 0.95$ (it prefers (1,0,0) at low SNR, the differencing (0,1,1) at higher SNR), but these approximate the ARMA(1,1) closely enough that the median innovation correlation with the Kalman filter stays at 0.99; forcing (1,0,1) restores 0.9995.

Appendix C. Summary of key quantities

Claim	Number
Raw CUSUM level 3σ detect, all SNRs	0.97–0.99
Innovation CUSUM delay vs raw, 3σ	24–53 vs 58–91 obs
Innovation CUSUM detect at 3σ	0.55–0.67
μ_∞ at 3σ, SNR 0.5 (knife-edge vs k=0.5)	0.469
r-break ×1.5: latent composite, T=500	0.82 / 0.87 / 0.91 (SNR 0.1/0.5/2.0)
r-break ×1.5: raw rung, T=500	1.00 / 0.56 / 0.10 (SNR 0.1/0.5/2.0)
r-break ×1.5: ARIMA rung, T=500	0.90 / 0.94 / 0.87 (SNR 0.1/0.5/2.0)
r-break ×1.5 t_5 (raw/ARIMA/composite), SNR 0.5	0.43 / 0.74 / 0.16
ARMA≡Kalman innovation $\bar{\rho}$ (estimated / true params)	0.99 / 1.000 (max|Δ|≈ 10^{-9})
q-break ×1.5: raw rung, T=500	0.09 / 0.21 / 0.23 (SNR 0.1/0.5/2.0)
q-break ×1.5: ARIMA rung, T=500	0.03 / 0.10 / 0.16 (SNR 0.1/0.5/2.0)
Pre-registered decision rule (§5) resolved	Outcome B2 (r-channel-specific)
φ sweep: μ_∞ sorts innovation-CUSUM detection	Spearman 0.9415, n=24 (4 φ × 3 SNR × 2 shifts); stratified permutation null (shuffle φ-pairing within SNR×shift, n_perm=20,000
Innovation CUSUM 3σ escape (φ=0.5 / 0.99)	0.98–1.00 / 0.30–0.63
Local-level 3σ level detect (all methods)	≤ 0.15 (≈ FAR)
Local-level ×1.5 variance (raw / ARIMA rung)	0.06 / 0.58–0.84
ARL_0 at 5% window-FAR (L=375)	≈ 7300 obs
φ×q: ×1.5 raw edge, φ-swept = SNR-swept	Δ 0.11=0.11 (SNR 0.5)
φ×q: subtle Δ at φ→ 0 / Spearman(amp,Δ)	0.00 / 0.83; ×3 falsified (−0.60)
Variance ×1.5 across T = 200/500/2000	0.11 / 0.87 / 0.99
t_5 collapse and repair (×1.5)	0.16 → 0.75 (tail_cusum)
Quieting ×⅔ (only tail_cusum)	0.41 / 0.33
Persistence-down, best anywhere	0.33 (SNR 2.0)
Multi-break: raw second-event recall (level→level)	0.00
Composite level→var second event	0.60 (F1 0.63)
Windowed-CUSUM fix, level→level 2nd event (raw / innovation)	0.00→ 0.68 / 0.01→ 0.23
Windowed-CUSUM fix, level→var / var→var 2nd event	no improvement (≈ 0.00): mean-shift only
PELT localization at FAR-matched 5%, level 3σ	0.83–0.92 (vs. causal raw CUSUM 0.97–0.99)
PELT localization at FAR-matched 5%, variance ×1.5/×3	0.00–0.20 (vs. dedicated raw variance-CUSUM 0.10–1.00)
INDPRO permutation p (composite)	0.008 (uncorrected — does not survive Bonferroni/BH-FDR across the 19 tests in Table 6; §9)

Claim	Number
Circular-shift joint test, INDPRO (5 methods, total hits)	observed 7 vs. null mean 2.25, SD 1.80 (20,000 draws), $p \approx 0.023$ — does not survive Bonferroni across 4 series; §9
GARCH(1,1) benchmark vs. raw/ARIMA rungs, ×1.5, SNR 0.5/2.0 (n_reps=500)	r-channel: 0.098/0.096 (GARCH) vs. 0.56/0.10 (raw), 0.94/0.87 (ARIMA); q-channel: 0.066/0.098 (GARCH) vs. 0.21/0.23 (raw), 0.
Mixed-channel (raw+ARIMA run jointly, unknown channel), SNR 0.1/0.5/2.0	combined loses to single-better detector at every SNR: 0.493 vs 0.553 / 0.490 vs 0.560 / 0.457 vs 0.623
GFC real-time	2008-09 data, known 2008-12
COVID real-time	data 2020-03, ~2 mo before NBER